

Machine Learning Analysis Report

AI Job Role Archetypes - DACH Region Skill Clustering

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1. Overview

This report presents an unsupervised machine learning analysis of AI job postings across the DACH region (Germany, Austria and Switzerland), with the goal of discovering natural skill-based role archetypes that may differ from the official job titles assigned by employers. K-Means clustering was applied to a binary skill matrix derived from job descriptions collected via the Adzuna Jobs API.

2. Dataset Description

The dataset was created by collecting AI job postings from the Adzuna Jobs API across Germany, Austria and Switzerland using 31 targeted search queries spanning five role categories: AI Engineering, Data Analytics, AI Product and Project Management, AI Integration and AI Consulting. The initial collection produced 485 unique job postings stored in `adzuna_ai_jobs_dach.csv` across 15 columns. Because the Adzuna API truncates descriptions at approximately 500 characters, a second enrichment script fetched full job pages from source URLs, increasing mean description length from 500 to 3,402 characters. The final modeling dataset comprises 304 job postings with at least one detected skill across two files: `adzuna_ai_jobs_dach_enriched.csv` (17 columns) and `adzuna_ai_skills_dach_enriched.csv` (4 columns: `job_id`, `skill`, `country`, `role_category`). Key features used for modeling are the binary skill indicators derived from the enriched job descriptions, covering 50 skills across programming languages, ML frameworks, cloud platforms, AI domains and tooling practices. There is no target variable - this is an unsupervised task.

3. Modeling Approach

Data preparation

Job postings with zero detected skills were removed, leaving 304 jobs for modeling. Each job was represented as a binary vector using `MultiLabelBinarizer` from `scikit-learn`

(Pedregosa et al., 2011), producing a 304 x 50 feature matrix where each dimension indicates the presence or absence of a skill. No scaling or normalisation was applied - binary features are already on a uniform scale and scaling would distort the equal weighting of skills. Metadata columns (title, country, role_category) were attached to the matrix after clustering for interpretability but excluded from the algorithm.

Model selection

K-Means clustering was selected as the modeling approach because it is well-suited to binary feature matrices, produces interpretable cluster centroids that reveal dominant skills per group, and scales efficiently to the dataset size. K-Means, originally described by MacQueen (1967), partitions data into k groups by minimising within-cluster sum of squared distances. The number of clusters k was determined using the elbow method, plotting inertia across $k = 2$ to $k = 10$, with $k = 5$ selected at the point of diminishing returns. The model was trained with $n_init = 10$ to reduce sensitivity to random initialisation.

Evaluation metrics

The silhouette score was used as the primary quantitative metric. Introduced by Rousseeuw (1987), it measures how similar each data point is to its own cluster compared to other clusters, ranging from -1 (worst) to +1 (best). For unsupervised tasks on real-world overlapping data, scores in the range 0.1–0.3 are typical and do not invalidate the clustering (Rousseeuw, 1987). Qualitative evaluation was conducted by inspecting cluster centroids and cross-tabulating cluster assignments against the human-assigned role categories.

Assumptions

K-Means assumes clusters are approximately spherical and of similar size and uses Euclidean distance as its similarity measure. These assumptions are imperfect for binary data, where cosine similarity would be theoretically more appropriate. The equal weighting of all skills also assumes no skill is inherently more discriminative than another, which may not reflect true role differentiation.

4. Results

The final K-Means model ($k=5$) produced five clusters of sizes 116, 38, 35, 76, and 39 jobs respectively. The silhouette score reflects moderate overlap between clusters, consistent with the genuine skill overlap present in AI job roles. The elbow and silhouette score plots are included in the notebook as viz_elbow.png and viz_silhouette.png.

Cluster interpretation via centroid analysis identified five distinct role archetypes:

(0) Data and Analytics Generalist - agile, machine learning, python, sql, git;
(1) Cloud ML Engineer - azure, python, gcp, aws, docker;
(2) MLOps and GenAI Engineer - python, machine learning, generative ai, kubernetes, mlops;
(3) AI Automation and Integration - scala, rpa, python, generative ai, aws;
(4) Deep Learning and AI Research - machine learning, python, computer vision, deep learning, generative ai. Cluster profiles are visualised in viz_skill_heatmap.png and cluster separation is shown in viz_clusters_pca.png.

Cross-tabulation of clusters against human-assigned role categories revealed that AI_Engineering dominates three of five clusters (45%, 71%, and 87%), while AI_Consulting and AI_PM are distributed thinly across all clusters. Cluster 0 - the largest at 116 jobs - is the most mixed, with Data_Analytics as the plurality at 41%.

5. Interpretation for a Non-Technical Audience

This analysis asked a simple question: do AI professionals in Germany, Austria and Switzerland actually divide into distinct groups based on the skills their jobs require or do the official job titles obscure more than they reveal? The short answer is: the skill-based groupings are real, but they do not always match the job titles employers use. Technical roles - particularly those focused on deep learning, cloud engineering, and MLOps - form the clearest and most distinct groups. Consulting and project management roles, by contrast, do not form their own skill-based clusters at all. This suggests that in practice, AI consultants and AI project managers in the DACH region are expected to have broadly the same technical skills as engineers, rather than a separate, business-focused skill profile. For job seekers, this means that the boundaries between role titles are more fluid than they appear and that building a strong technical foundation is relevant across a wider range of job titles than one might expect.

6. Limitations and Potential Bias

Dataset limitation

The Adzuna API truncates job descriptions at approximately 500 characters and 155 of 485 postings (32%) could not be enriched via the secondary fetch step due to JavaScript-rendered pages. These jobs were excluded from modeling, introducing a potential selection bias - roles hosted on modern, JavaScript-heavy career platforms may be systematically underrepresented. Additionally, Austria (34 postings) and Switzerland (41 postings) are heavily underrepresented relative to Germany (410 postings), meaning the clusters primarily reflect the German AI job market rather than the DACH region as a whole.

Potential source of bias

The skill taxonomy used for extraction was manually defined in English with selected German variants. Skills expressed in German terminology not covered by the taxonomy will have been missed, potentially undercounting the true skill diversity of postings from Austrian and Swiss employers who write in German. Furthermore, K-Means assumes equal skill weighting and spherical cluster geometry, which may not reflect the true structure of the job market. As Pedregosa et al. (2011) note, the choice of algorithm and distance metric materially affects clustering outcomes, and results should be interpreted as one possible partitioning rather than a definitive taxonomy.

7. References

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